
Session 1 — Probability I

What today is about

Almost everything we study in cognitive science is *uncertain*: the next response of a participant, the number of spikes a neuron fires, whether a model gets an answer right.

Probability is simply the careful mathematics of that uncertainty. Today we build it from the ground up.

- Describe what can happen: *sample spaces* and *events*.
- Combine chances with a few simple *rules*.
- Update a belief when new evidence arrives: **Bayes' theorem**.
- Meet the handful of *distributions* that describe most real data.

The one idea to hold onto

Everything today leads to one destination.

If you remember only one thing

Bayes' theorem tells us how observing evidence should change what we believe. It is the mathematical backbone of Bayesian models of perception, learning, and decision-making — and of machine learning.

We will spend the first hour building the tools we need to understand it, and the second hour using it and meeting the common distributions.

A probability is just a number between 0 and 1

Before any formulas, fix the picture in your head.

- A probability measures our *degree of certainty* that something happens.
- 0 means impossible, 1 means certain, 0.5 means “as likely as not”.
- Everything else is rules for combining these numbers without contradicting ourselves.

Intuition

Think of one unit of “belief” that we spread across all the things that could happen. Probability is just how much of that belief sits on each outcome.

Why this matters for cognitive science

This is not only mathematics — it is a theory of the mind.

A leading view in modern cognitive science is that the brain is, in part, a *probabilistic inference machine*. Perception is treated as inferring the most probable cause of ambiguous sensory data; learning as updating beliefs from experience.

Link to cognitive science

Bayesian models of cognition, ideal-observer analysis, and the probabilistic models underlying machine learning all speak the language we build today. Learning it lets you read a large part of the modern literature.

Step 1: list everything that can happen

To reason about chance we first say, precisely, what the possible outcomes are.

Definition

The **sample space** Ω is the set of all possible outcomes of an experiment. An **event** A is a subset of Ω — a collection of outcomes we care about.

A few concrete examples:

- One die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$; the event “even” is $\{2, 4, 6\}$.
- A detection task: $\Omega = \{\text{yes}, \text{no}\}$.
- Spikes in one second: $\Omega = \{0, 1, 2, 3, \dots\}$.

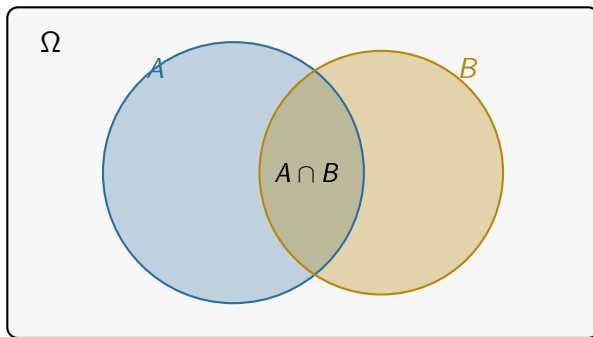
Events are yes/no questions about the outcome

An event is just a question with a yes/no answer: “was the die even?” splits Ω into the outcomes where the answer is yes and those where it is no.

Because events are sets, we combine them with the same words we use in ordinary logic:

- $A \cup B$ — “*A or B*”: outcomes in either (or both).
- $A \cap B$ — “*A and B*”: outcomes in both at once.
- A^c — “*not A*”: every outcome outside A .
- A, B are *disjoint* if they cannot both happen: $A \cap B = \emptyset$.

Picturing events: the Venn diagram



The whole rectangle is Ω . The overlap is “ A and B ”; the two circles together are “ A or B ”; everything outside both circles is “neither”.

When outcomes are equally likely, we just count

If every outcome is equally likely, probability becomes a counting problem:

$$\mathbb{P}(A) = \frac{\# \text{ outcomes in } A}{\# \text{ outcomes in } \Omega}.$$

So the skill we need is counting outcomes efficiently. Two ideas cover almost everything.

Permutations vs. combinations

Order matters (permutations): $\frac{n!}{(n-k)!}$ ways to arrange k of n .

Order does not (combinations): $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ ways to choose k of n .

A reading group and a podium

A lab has 8 members.

How many 3-person reading groups? A group is a *set* — order does not matter — so we use a combination:

$$\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56.$$

How many gold–silver–bronze podiums? Now order *does* matter, so we use a permutation: $8 \cdot 7 \cdot 6 = 336$.

The podium counts each trio in all $3! = 6$ possible orders — which is exactly why the group count is $336/6 = 56$.

Practice — counting

Try these on your own; we will solve them together on the board.

Practice — try it

1. A questionnaire has 5 yes/no items. How many different response patterns are possible?
2. From a group of 10 participants, how many ways are there to form a 4-person focus group?
3. In how many different orders can 6 pictures be shown to a subject?

Three common-sense rules (the axioms)

Any sensible way of assigning probabilities must obey three rules. They simply formalise common sense.

Kolmogorov's axioms

- (1) $\mathbb{P}(A) \geq 0$ — no negative chances.
- (2) $\mathbb{P}(\Omega) = 1$ — something must happen.
- (3) If A and B are disjoint, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Everything else in probability can be derived from just these three statements.

Two rules you will use constantly

From the axioms come the two everyday tools.

Useful fact

Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Addition: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Why the subtraction? If we simply add $\mathbb{P}(A)$ and $\mathbb{P}(B)$, we count the overlap $A \cap B$ *twice* — once in each. Subtracting it once puts things right. If the events are disjoint there is no overlap, and the rule reduces to plain addition.

Addition rule — worked example

Example

In a study, 60% of participants are bilingual (B), 30% are musicians (M), and 20% are both.

Reading the answers straight off the rules:

- *Bilingual or musician*: $\mathbb{P}(B \cup M) = 0.60 + 0.30 - 0.20 = 0.70$.
- *Neither*: by the complement rule, $1 - 0.70 = 0.30$.
- *Bilingual but not a musician*: $\mathbb{P}(B) - \mathbb{P}(B \cap M) = 0.60 - 0.20 = 0.40$.

A trick worth its own slide: “at least one”

Questions about “at least one” are almost always easier through the *complement*: it is usually simpler to compute the chance of *none*.

Example

A target appears on each of 3 independent trials with probability 0.2.

The chance of *no* target on a single trial is 0.8, so across three independent trials

$$\mathbb{P}(\text{no target at all}) = 0.8^3 = 0.512,$$

and therefore $\mathbb{P}(\text{at least one target}) = 1 - 0.512 = 0.488$. Much easier than adding up the “exactly one / exactly two / exactly three” cases.

A pitfall to flag early

Common pitfall

The addition rule collapses to $\mathbb{P}(A) + \mathbb{P}(B)$ *only when the events are disjoint*. Forgetting the $-\mathbb{P}(A \cap B)$ term is the single most common early mistake.

A useful self-check: if a probability calculation ever gives you a number greater than 1, you have almost certainly double-counted an overlap somewhere. A probability can never exceed 1.

Practice — try it

In a sample, $\mathbb{P}(\text{coffee}) = 0.5$, $\mathbb{P}(\text{tea}) = 0.3$, and $\mathbb{P}(\text{both}) = 0.1$. Find:

1. $\mathbb{P}(\text{coffee or tea})$;
2. $\mathbb{P}(\text{neither})$;
3. $\mathbb{P}(\text{exactly one of the two})$.

Learning from information: conditional probability

The central question of the whole session: *how does knowing one thing change the probability of another?*

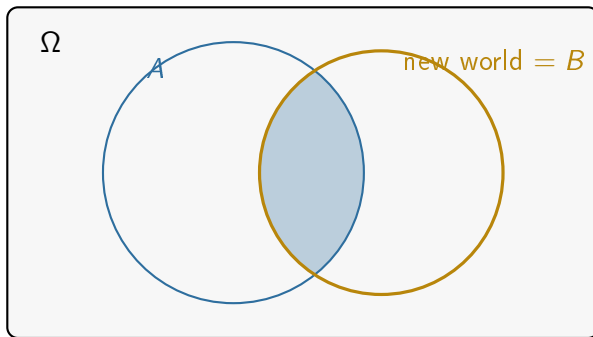
Definition

The **conditional probability** of A given B (with $\mathbb{P}(B) > 0$) is

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Read it as: “among the cases where B happened, how often does A also happen?”

Conditioning means zooming in



Conditioning on B throws away everything outside the B circle. Our new “world” is B , and $\mathbb{P}(A \mid B)$ is the fraction of that world covered by A . Dividing by $\mathbb{P}(B)$ rescales it so the shrunken world again totals 1.

The multiplication rule, and a quick example

Rearranging the definition gives a way to build the probability of “ A and B ” step by step:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \mathbb{P}(B).$$

Example

Roll a fair die. What is $\mathbb{P}(\text{even} \mid \text{greater than } 3)$?

The condition “greater than 3” leaves the outcomes $\{4, 5, 6\}$. Among those, the even ones are $\{4, 6\}$. So

$$\mathbb{P}(\text{even} \mid > 3) = \frac{2}{3},$$

up from $\frac{1}{2}$ without the information. Knowing more changed the probability.

Averaging over scenarios: total probability

Often we know how likely A is *within* each of several scenarios, and want its overall probability.

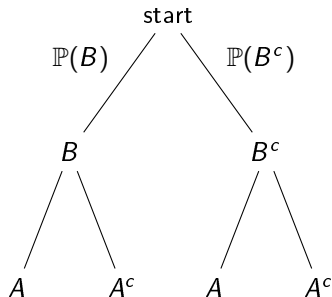
Law of total probability

If $B_1, \dots, B_k$ are disjoint and cover everything, then

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

In words: work out how likely A is in each scenario, then take a weighted average, weighting each scenario by how likely it is. A picture makes this automatic — see the next slide.

The probability tree



Multiply probabilities along a branch, then *add* across the branches that end in A :

$$\mathbb{P}(A) = \mathbb{P}(B)\mathbb{P}(A \mid B) + \mathbb{P}(B^c)\mathbb{P}(A \mid B^c).$$

Total probability — worked example

Two boxes

Box 1 holds 3 red and 1 blue token; Box 2 holds 1 red and 3 blue. You pick a box at random (each with probability $\frac{1}{2}$) and draw one token.

What is $\mathbb{P}(\text{red})$? Average the two scenarios:

$$\mathbb{P}(\text{red}) = \underbrace{\frac{3}{4} \cdot \frac{1}{2}}_{\text{Box 1}} + \underbrace{\frac{1}{4} \cdot \frac{1}{2}}_{\text{Box 2}} = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}.$$

Keep this number in mind: next we will *reverse* the question — “given a red token, which box was it probably from?” — and this $\frac{1}{2}$ becomes the denominator.

Independence: when information tells you nothing

Definition

A and B are **independent** if knowing one tells you nothing about the other:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A), \quad \text{equivalently} \quad \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Independence is what lets us multiply probabilities of separate trials — the coin does not remember its last flip. It is an assumption we make constantly, and it is usually an *approximation*.

Independent is not the same as disjoint

Common pitfall

Independent and *disjoint* are almost opposites. Disjoint events cannot both happen, so learning that one occurred tells you the other did *not* — that is maximal *dependence*. Independence means the events carry *no* information about each other.

Students conflate these two constantly. If in doubt, go back to the definition: independence is about $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, not about the events being mutually exclusive.

Why independence matters in modeling

Link to cognitive science

Independence assumptions are what make cognitive and machine-learning models computable. The “naive Bayes” classifier assumes features are independent given the category; many models of memory and categorisation assume trials are independent.

These assumptions are rarely exactly true, but they are often good enough — and a large part of the modeler’s craft is knowing *when* they break down.

Bayes' theorem: updating a belief

Now the destination. Bayes' theorem flips a conditional probability around, which is exactly what “learning from evidence” requires.

Key idea

$$\mathbb{P}(H | E) = \frac{\mathbb{P}(E | H) \mathbb{P}(H)}{\mathbb{P}(E)} \propto \mathbb{P}(E | H) \mathbb{P}(H),$$

where H is a hypothesis and E is the evidence we observe.

The next slide names the four pieces — each has a job worth learning.

The four pieces of Bayes' theorem

- **Prior** $\mathbb{P}(H)$ — what we believed *before* the evidence.
- **Likelihood** $\mathbb{P}(E \mid H)$ — how well the hypothesis *predicts* the evidence.
- **Posterior** $\mathbb{P}(H \mid E)$ — our *updated* belief.
- **Evidence** $\mathbb{P}(E)$ — a normaliser, from the law of total probability:
$$\mathbb{P}(E) = \mathbb{P}(E \mid H)\mathbb{P}(H) + \mathbb{P}(E \mid H^c)\mathbb{P}(H^c).$$

Intuition

New belief $\propto$ (how well the hypothesis predicted what we saw) $\times$ (how much we believed it beforehand).

The classic medical test — the setup

Example

A disease affects 1% of people. A test correctly flags 99% of true cases (sensitivity), but also wrongly flags 5% of healthy people (false positives). A randomly chosen person tests positive.

The natural question — “they tested positive, so they probably have it, right?” — turns out to be badly wrong. Let us see why, first with the formula, then with an even clearer picture.

The medical test — the calculation

Write D for “has the disease” and $+$ for “tests positive”. First the denominator, the overall chance of a positive test:

$$\mathbb{P}(+) = \underbrace{0.99 \times 0.01}_{\text{sick} = 0.0099} + \underbrace{0.05 \times 0.99}_{\text{healthy} = 0.0495} = 0.0594.$$

Now Bayes:

$$\mathbb{P}(D | +) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} = \frac{1}{6} \approx 16.7\%.$$

A positive result means only about a 1-in-6 chance of disease.

The medical test — why, in plain numbers

The cleanest way to see it: imagine 10,000 people.

- 100 are sick; the test catches 99 of them.
- 9,900 are healthy; the test wrongly flags 5%, that is 495 of them.
- So $99 + 495 = 594$ people test positive, but only 99 are actually sick.

$$\mathbb{P}(\text{sick} \mid +) = \frac{99}{99 + 495} = \frac{1}{6}.$$

Because the disease is *rare*, the many false positives from the huge healthy group swamp the few true positives.

This is a famous cognitive bias

Link to cognitive science

People — including doctors — systematically overestimate $\mathbb{P}(D \mid +)$ because they focus on the test's accuracy and neglect the low base rate $\mathbb{P}(D)$. This is the **base-rate fallacy**, one of the most robust findings in the psychology of judgment (Kahneman & Tversky).

The very same mathematics describes Bayesian models of perception: the “prior” is the brain's expectation, and the “likelihood” is the incoming sensory evidence.

A shortcut: Bayes in odds form

Sometimes it is quicker to work with *odds* and skip the denominator entirely.

Useful fact

$$\underbrace{\frac{\mathbb{P}(H | E)}{\mathbb{P}(H^c | E)}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(E | H)}{\mathbb{P}(E | H^c)}}_{\text{likelihood ratio}} \times \underbrace{\frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}}_{\text{prior odds}}.$$

Evidence simply *multiplies* the odds. For the medical test: prior odds 1:99, likelihood ratio $0.99/0.05 \approx 19.8$, giving posterior odds about 1:5 — the same 1/6 as before.

Practice — try it

A jar of candies is either *ordinary* (30% red) or *special* (70% red); beforehand each is equally likely. You draw one candy and it is red.

1. What is $\mathbb{P}(\text{red})$ overall?
2. What is $\mathbb{P}(\text{special} \mid \text{red})$?
3. Redo part (2) using prior odds $\times$ likelihood ratio.

From outcomes to numbers: random variables

So far outcomes have been labels (heads, even, positive). Usually we want *numbers* we can average and plot.

Definition

A **random variable** X assigns a number to every outcome. It is *discrete* if it takes separate values $(0, 1, 2, \dots)$ and *continuous* if it can take any value in a range.

Examples: the number of correct trials out of 20 (discrete); a reaction time in milliseconds (continuous); a spike count (discrete); a voltage (continuous).

Where a random variable's probability “lives”

Two functions describe a random variable, and one more accumulates it.

Useful fact

Discrete — the *probability mass function* $p(x) = \mathbb{P}(X = x)$; the masses are ≥ 0 and sum to 1.

Continuous — the *probability density function* $f(x)$; areas under it give probabilities, and the total area is 1.

Either — the *cumulative distribution function* $F(x) = \mathbb{P}(X \leq x)$, rising from 0 to 1.

A mass function is a bar chart; a density is the smooth curve you get as the bars become infinitely thin.

A subtlety worth stressing

Common pitfall

For a continuous variable, $f(x)$ is a *density*, not a probability. It can be larger than 1, and the probability of any *single exact* value is zero. Only *intervals* (areas) carry probability.

So “the probability the reaction time is exactly 300.000... ms” is zero, but “between 290 and 310 ms” is a real, positive number — the area under the curve across that interval.

Practice — try it

A continuous variable has density $f(x) = 2x$ for $0 \leq x \leq 1$, and 0 elsewhere.

1. Check that this is a valid density.
2. Find $\mathbb{P}(X \leq 0.5)$.
3. Explain why $\mathbb{P}(X = 0.5) = 0$.

Four distributions cover most of what you will meet

A “distribution” is a named, reusable shape of randomness. We will meet four, each with its own story — learn the story, not just the formula.

- **Bernoulli** — a single yes/no trial.
- **Binomial** — counting successes across many trials.
- **Poisson** — counts of rare events (e.g. spikes).
- **Gaussian** — the bell curve of noise and averages.

Bernoulli and Binomial

Bernoulli(p) is one yes/no trial: success with probability p . It is the atom of all binary data — one coin flip, one correct/incorrect response.

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

Binomial(n, p) counts the successes in n independent Bernoulli trials:

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad \mathbb{E}[X] = np.$$

So a Binomial is just a *sum* of n Bernoullis.

Guessing on a multiple-choice test

A student guesses on 10 four-option questions, so $p = \frac{1}{4}$ on each.

The probability of exactly 3 correct:

$$\mathbb{P}(X = 3) = \binom{10}{3} (0.25)^3 (0.75)^7 \approx 0.250.$$

The expected number correct is $\mathbb{E}[X] = np = 10 \times 0.25 = 2.5$. This is why pure guessing on a 10-question quiz typically yields about 2–3 right.

Poisson — counts of rare events

When events happen independently at a steady average rate, the number in a fixed window follows a **Poisson**(λ), where λ is *both* the mean and the variance:

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \mathbb{E}[X] = \text{Var}(X) = \lambda.$$

Link to cognitive science

This is the standard first model of **neural spike counts**. Its signature, “variance = mean”, is used all the time to check whether real spike data are more or less variable than this baseline.

Example

A neuron fires on average $\lambda = 5$ spikes per second. What is the probability of observing exactly 3 spikes in a given second?

$$\mathbb{P}(X = 3) = \frac{5^3 e^{-5}}{3!} = \frac{125 \times 0.006738}{6} \approx 0.140.$$

So even though the *average* is 5, seeing only 3 spikes in a particular second is entirely ordinary — about a 14% chance. Spike counts are genuinely noisy.

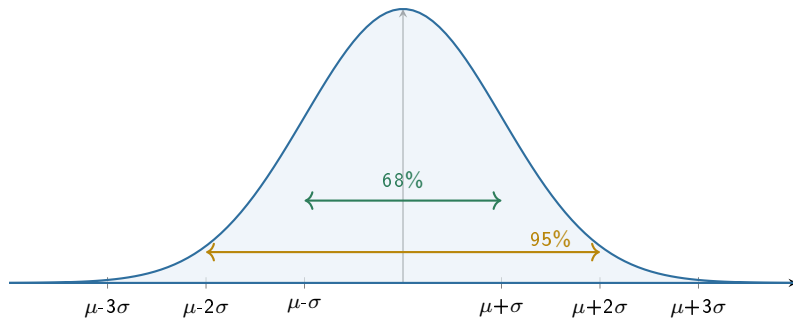
The Gaussian: the bell curve

The **Normal** (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$ is the continuous workhorse. It is symmetric and bell-shaped, centred at the mean μ , with its width set by the standard deviation σ :

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

The next slide shows the one fact about it that lets you reason without ever doing an integral.

The 68–95–99.7 rule



About 68% of the probability lies within $\pm 1\sigma$ of the mean, 95% within $\pm 2\sigma$, and 99.7% within $\pm 3\sigma$ — for every Normal distribution.

Reading off Normal probabilities

To use the rule for any Normal, we *standardise*: measure how many standard deviations a value is from the mean, $z = \frac{x - \mu}{\sigma}$.

Example

Adult heights are roughly $\mathcal{N}(\mu = 170, \sigma = 10)$ cm. What fraction of people are taller than 185 cm?

Standardise: $z = \frac{185 - 170}{10} = 1.5$. From a z-table (or the rule), $\mathbb{P}(Z > 1.5) \approx 0.067$ — about 6.7% of people exceed 185 cm.

The four distributions at a glance

Distribution	Models	Support	Mean	Variance
Bernoulli(p)	one yes/no trial	$\{0, 1\}$	p	$p(1 - p)$
Binomial(n, p)	# successes in n	$\{0, \dots, n\}$	np	$np(1 - p)$
Poisson(λ)	counts of rare events	$\{0, 1, 2, \dots\}$	λ	λ
Normal(μ, σ^2)	noise / averages	$\mathbb{R}$	μ	σ^2

Two links: a Binomial is a sum of Bernoullis; and for large n a Binomial looks Normal (a preview of the Central Limit Theorem).

Why the bell curve is everywhere

Link to cognitive science

Whenever a quantity is the sum of many small, independent influences, its distribution is approximately Normal — regardless of the details. This is the **Central Limit Theorem**, which we will make intuitive next session.

It explains why test scores, measurement errors, and averaged neural signals all look bell-shaped, and why Gaussian noise is the default assumption in almost every model of perception and data analysis.

Practice — try it

1. A subject responds correctly on 70% of trials, independently. Over 8 trials, what is the expected number correct, and $\mathbb{P}(\text{all 8 correct})$?
2. A neuron fires at $\lambda = 2$ spikes per window. What is $\mathbb{P}(X = 0)$?
3. Reaction times are $\mathcal{N}(500, 50^2)$ ms. Roughly what fraction fall between 450 and 550 ms?

Session 1 — what we built

- Probabilities are numbers in $[0, 1]$ obeying the complement and addition rules.
- **Conditional probability** formalises learning from information; *independence* is its absence.
- **Bayes' theorem** turns evidence into updated belief — the engine of Bayesian cognition and the base-rate fallacy.
- **Random variables** and four *distributions* describe binary trials, counts, rare events, and bell-shaped noise.

Next session: from single quantities to *data* — summarising it and inferring from it.